

HDOC – Experiment 36

Question

Write a menu-driven C program to perform the following operations using separate functions:

1. Count characters in a string without using built-in length functions.
2. Print each character of a string on a new line.
3. Count vowels and consonants in a string.
4. Exit.

❖ Algorithm

1. Start.
2. Declare a character array str to store the string.
3. Read a string from the user using fgets().
4. Remove the newline character added by fgets() manually.
5. Display the menu:
 - 1. Count characters in the string.
 - 2. Print each character on a new line.
 - 3. Count vowels and consonants.
 - 4. Exit.
6. Read the user's choice.

7. Use a switch statement:

- **Choice 1:** Call countCharacters() and display the number of characters.
- **Choice 2:** Call printCharacters() to display every character on a separate line.
- **Choice 3:** Call countVowelsConsonants() to count and display vowels and consonants.
- **Choice 4:** Exit the program.
- Otherwise, display an invalid-choice message.

8. Repeat the menu until the user selects Exit.

9. Stop.

❖ Program code in C

```
#include <stdio.h>
```

```
// Function prototypes
```

```
int countCharacters(char str[]);
```

```
void printCharacters(char str[]);
```

```
void countVowelsConsonants(char str[]);
```

```
int main()
```

```
{
```

```
    char str[200];
```

```
    int choice, i;
```

```
    printf("Enter a string: ");
```

```
    fgets(str, sizeof(str), stdin);
```

```
// Remove newline character added by fgets()
for (i = 0; str[i] != '\0'; i++)
{
    if (str[i] == '\n')
    {
        str[i] = '\0';
        break;
    }
}

do
{
    printf("\n----- STRING OPERATIONS ----- \n");
    printf("1. Count characters in the string\n");
    printf("2. Print each character on a new line\n");
    printf("3. Count vowels and consonants\n");
    printf("4. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            printf("Number of characters = %d\n",
                countCharacters(str));
            break;
```

case 2:

```
printf("Characters in the string:\n");
```

```
printCharacters(str);
```

```
break;
```

case 3:

```
countVowelsConsonants(str);
```

```
break;
```

case 4:

```
printf("Exiting the program...\n");
```

```
break;
```

default:

```
printf("Invalid choice! Please enter 1 to 4.\n");
```

```
}
```

```
} while (choice != 4);
```

```
return 0;
```

```
}
```

```
// Function to count characters without strlen()
```

```
int countCharacters(char str[])
```

```
{
```

```
int count = 0;
```

```
while (str[count] != '\0')
{
    count++;
}

return count;
}
```

// Function to print each character on a new line

```
void printCharacters(char str[])
```

```
{
    int i = 0;

    while (str[i] != '\0')
    {
        printf("%c\n", str[i]);
        i++;
    }
}
```

// Function to count vowels and consonants

```
void countVowelsConsonants(char str[])
```

```
{
    int i = 0;

    int vowels = 0, consonants = 0;

    char ch;

    while (str[i] != '\0')
```

```
{  
    ch = str[i];  
  
    // Check whether the character is an alphabet  
    if ((ch >= 'A' && ch <= 'Z') ||  
        (ch >= 'a' && ch <= 'z'))  
    {  
        // Check whether it is a vowel  
        if (ch == 'A' || ch == 'E' || ch == 'I' ||  
            ch == 'O' || ch == 'U' ||  
            ch == 'a' || ch == 'e' || ch == 'i' ||  
            ch == 'o' || ch == 'u')  
        {  
            vowels++;  
        }  
        else  
        {  
            consonants++;  
        }  
    }  
  
    i++;  
}  
  
printf("Number of vowels = %d\n", vowels);  
printf("Number of consonants = %d\n", consonants);  
}
```

❖ Test case table

Test	Input String	Choice	Expected Result
1	Hello	1	Number of characters = 5
2	Hello	2	H, e, l, l, o printed on separate lines
3	Hello	3	Vowels = 2, Consonants = 3
4	Computer Science	1	Number of characters = 16 including the space
5	AEIOU	3	Vowels = 5, Consonants = 0
6	BCDF	3	Vowels = 0, Consonants = 4
7	CSE 123	3	Vowels = 1, Consonants = 2; spaces/digits ignored
8	Hello	5	Invalid choice! Please enter 1 to 4.
9	Hello	4	Program exits

❖ Compile, Execute and Evaluate

```
(kali㉿kali)-[~/Desktop/exp36]
$ vi string.c

(kali㉿kali)-[~/Desktop/exp36]
$ gcc string.c -o string

(kali㉿kali)-[~/Desktop/exp36]
$ ./string
Enter a string: Hello

—— STRING OPERATIONS ——
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 1
Number of characters = 5

—— STRING OPERATIONS ——
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 2
Characters in the string:
H
e
l
l
o

—— STRING OPERATIONS ——
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 3
Number of vowels = 2
Number of consonants = 3
```


—— STRING OPERATIONS ——

1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 4

Exiting the program...

(kali㉿kali)-[~/Desktop/exp36]

\$./string

Enter a string: Computer Science

—— STRING OPERATIONS ——

1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 1

Number of characters = 16

—— STRING OPERATIONS ——

1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 4

Exiting the program...

(kali㉿kali)-[~/Desktop/exp36]

\$./string

Enter a string: AEIOU

—— STRING OPERATIONS ——

1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 3

```
Number of vowels = 5
Number of consonants = 0
```

—— STRING OPERATIONS ——

1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 4

Exiting the program...

```
(kali㉿kali)-[~/Desktop/exp36]
```

```
$ ./string
```

Enter a string: BCD F

—— STRING OPERATIONS ——

1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 3

Number of vowels = 0

Number of consonants = 4

—— STRING OPERATIONS ——

1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 4

Exiting the program...

```
(kali㉿kali)-[~/Desktop/exp36]
```

```
$ ./string
```

Enter a string: CSE 123

—— STRING OPERATIONS ——

1. Count characters in the string
2. Print each character on a new line

1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 3

Number of vowels = 1

Number of consonants = 2

——— STRING OPERATIONS ———

1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 4

Exiting the program...

```
(kali㉿kali)-[~/Desktop/exp36]  
$ ./string
```

Enter a string: Hello

——— STRING OPERATIONS ———

1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 5

Invalid choice! Please enter 1 to 4.

——— STRING OPERATIONS ———

1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit

Enter your choice: 4

Exiting the program...

```
(kali㉿kali)-[~/Desktop/exp36]  
$
```